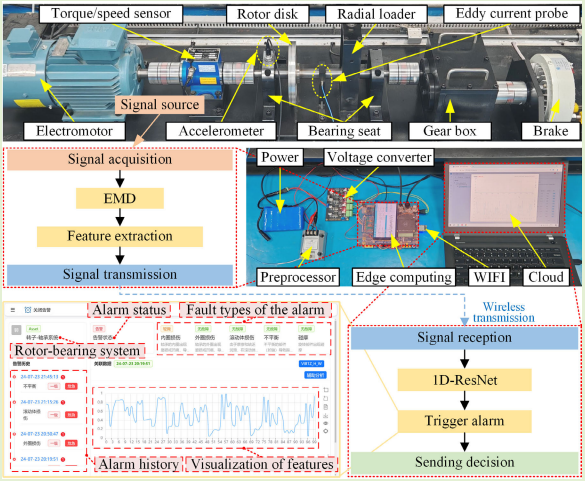
Cloud–Edge Collaborative Intelligent Fault Diagnosis of Rotor-Bearing System: Methodology and Experiment

Hongwei Fan, *Member, IEEE*, Haowen Xu, Wei Fan, Buran Chen, Qingshan Li, Teng Zhang, Xiangang Cao, and Xuhui Zhang, *Member, IEEE*

***Abstract*—The condition monitoring and fault diagnosis of rotor-bearing systems are crucial for rotating machinery. The traditional methods rely on either a single edge computing of limited data processing capabilities or a single cloud computing of large data transmission volumes. In this article, a novel monitoring and diagnosis framework for rotor-bearing systems is proposed by combining cloud and edge computing. First, a cloud–edge collaborative diagnosis architecture is designed, where data acquisition, signal processing, and feature extraction are assigned to the edge end, and a deep learning-based diagnosis task is assigned to the cloud end. The edge ends collect acceleration and displacement signals in real time through piezoelectric and eddy current sensors and uses empirical mode decomposition (EMD) and time–frequency feature extraction to realize preliminary feature expression of the obtained signals. Second, the feature data are sent to the cloud end from the edge end, where an improved 1-D residual network (1D-ResNet) is used to classify the feature data and realize data-driven fault diagnosis. To verify the effectiveness of the methodology, a cloud–edge collaborative fault diagnosis system is developed, including a rotor-bearing unit simulation platform, an edge-end condition monitoring device, and a cloud-end diagnosis and visualization platform. The results show that the cloud–edge collaborative fault diagnosis method and system excel in terms of accuracy and real-time performance, where the accuracy reaches 99.9% at 1400 r/min and the consuming time is about 1.4 s.**



***Index Terms*— Cloud–edge collaboration, empirical mode decomposition (EMD), fault diagnosis, feature extraction, residual network (ResNet), rotor-bearing system.**

I. Introduction

THE intelligent fault diagnosis of mechanical equipment has become an important means to ensure the safe and stable operation of a production system in the context of the fourth industrial revolution and the rapid development of artificial intelligence technology [1]. As the core component of rotating machinery, the operating status of a rotor-bearing system directly affects the performance and life of the entire equipment and may even cause serious safety accidents [2]. Thus, using artificial intelligence technology to realize mechanical fault diagnosis is of great significance to the safe production and reliable operation of industrial equipment.

The development of cloud computing and edge computing has provided new solutions for intelligent fault diagnosis. Cloud computing has powerful data storage and processing capabilities and can perform in-depth analysis of massive amounts of data [3]. It usually uploads the collected on-site equipment status data to a cloud server, which completes a large number of computing tasks and displays the results remotely or sends them to the site [4]. With the development of intelligent diagnosis technology, the number of on-site machines and sensors is quickly increasing. The process of

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uploading original data to the cloud end will generate PB-level or even ZB-level data capacity, which can easily lead to some problems such as high latency, large data loss, and serious network blockage [5]. As a supplement to cloud computing, edge computing can process original data in a timely manner near the machine, reducing data transmission delays and bandwidth consumption [6]. However, the limited computing ability of edge devices sometimes makes it difficult to support complex data processing [7]. The cloud–edge collaborative scheme combines the advantages of both and provides a more scientific and efficient solution for intelligent fault diagnosis [8]. Xiang et al. [9], in order to solve the problem of difficult and time-consuming fault diagnosis of the transformer, extracted its features using principal component analysis (PCA) at the edge end, and deployed extreme gradient boosting (XGBoost) model at the cloud end for state identification, which realizes the transformer monitoring and fault diagnosis based on cloud–edge collaboration. Chen et al. [10] proposed a cloud–edge collaborative deep learning scheme for infrared thermal images to carry out intelligent fault diagnosis of electrical equipment. Ren et al. [11] proposed a cloud–edge collaborative monitoring and diagnostic method to deal with the problems of fault data missing and imbalance, which proved the effectiveness of the proposed method. Sima et al. [12] proposed a cloud–edge collaborative method based on transfer learning and federated learning, which solves the problem of fault diagnosis of power lines. In summary, cloud–edge collaboration is an effective manner for equipment fault diagnosis.

For rotor-bearing system fault diagnosis, when a data-driven approach is used [13], the equipment status can be characterized by different signals, such as vibration [14], infrared thermal imaging [15], and sound [16], and the fault diagnosis can be achieved by analyzing and processing these signals or using the machine learning method. In recent years, with the gradual application of deep learning in the field of mechanical fault diagnosis [17], it is expected to improve the fault diagnosis capability of mechanical equipment. Sun et al. [18] proposed a bearing fault diagnosis method based on empirical mode decomposition (EMD) and improved Chebyshev distance. Shao et al. [19] proposed a transfer learning and improved convolutional neural network (CNN)-based fault diagnosis method for rotor-bearing systems under variable conditions, in which the infrared thermal image was used to characterize the health condition of the equipment. Wang et al. [20] proposed a fault diagnosis method based on variational mode decomposition (VMD)-Hilbert spectrum and ShuffleNet-V2, which solves the problem of difficulty in extracting the fault characteristics of gears affected by coal flow load and power frequency. Zeng et al. [21] proposed a fault diagnosis method for belt conveyor rolling bearings based on CGM-MobileNet, which overcomes the diagnostic challenges of rolling bearings under various operating conditions and high noise interference. Xin et al. [22] proposed an infrared thermal image-based improved Gaussian convolutional deep belief network (GCDBN) for intelligent fault diagnosis of rotor-bearing systems at variable speeds. Hongwei et al. [23] used an information fused EMD-polar angle adaptive distribution of vibration images with CNN and support vector machine (SVM) for feature extraction and data classification to realize compound fault diagnosis of a rotor-bearing system. Yuan et al. [24] proposed a fault diagnosis method for multichannel rotor-bearing systems based on multistructure collaborative discriminant embedding, which improves the performance of the fault diagnosis model. In addition, Fan et al. [25], [26] converted the vibration signals into grayscale texture images, and investigated the intelligent fault diagnosis method for the rotor-bearing system of variable duty motors in terms of improving the model generalization ability and lightweight. In summary, it can be seen that signal processing combined with deep learning methods can realize the accurate fault diagnosis of mechanical equipment.

The cloud–edge collaboration provides a new solution for fault diagnosis of mechanical equipment, which utilizes the advantages of edge computing and cloud computing, respectively, to make the diagnosis process more accurate and efficient. In this article, a cloud–edge collaborative intelligent fault diagnosis framework is proposed for rotor-bearing systems. EMD-based vibration signal analysis with time–frequency feature extraction and improved 1-D residual network (1D-ResNet) is investigated, which reduces the transmission amount of data between the edge end and the cloud end while ensuring accuracy and improving efficiency. On this basis, a set of edge computing and cloud service platforms suitable for online monitoring and fault diagnosis is developed, and the corresponding algorithms are integrated into the edge device and cloud server to realize cloud–edge collaborative intelligent fault diagnosis of the rotor-bearing system.

This article is organized as follows. Section II builds the cloud–edge collaborative fault diagnosis framework. Section III introduces the algorithms and datasets used in this work. Section IV performs the experimental validation of fault diagnosis algorithms. Section V performs the deployment of the algorithms and system test, and finally, the conclusion is provided in Section VI.

II. Framework of Cloud–Edge Collaborative Fault Diagnosis

The cloud–edge collaborative platform designed in this study is shown in Fig. 1. The edge end is responsible for collecting data, executing signal processing, and transmitting data to the cloud end. The cloud end is responsible for managing the data from the edge end, executing the fault diagnosis, and sending the results to the site. The cloud and edge work together to accomplish the online fault diagnosis task of a rotor-bearing system. The solution can make full use of distributed computing resources to achieve localized processing of monitored data and reduce the burden of cloud computing while improving data security and privacy.

A. Edge-End Scheme

As shown in Fig. 1, the hardware devices [6] at the edge end mainly include sensors, conditioning circuits, A/D

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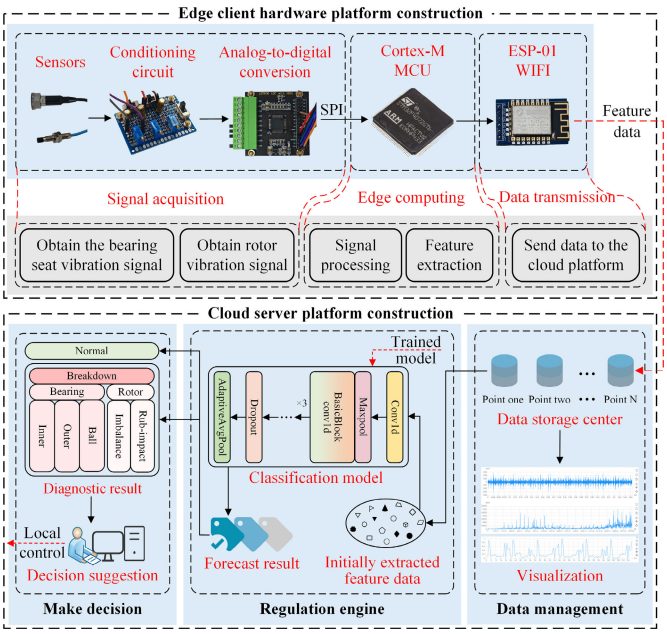


Fig. 1. Cloud–edge collaborative framework of fault diagnosis.

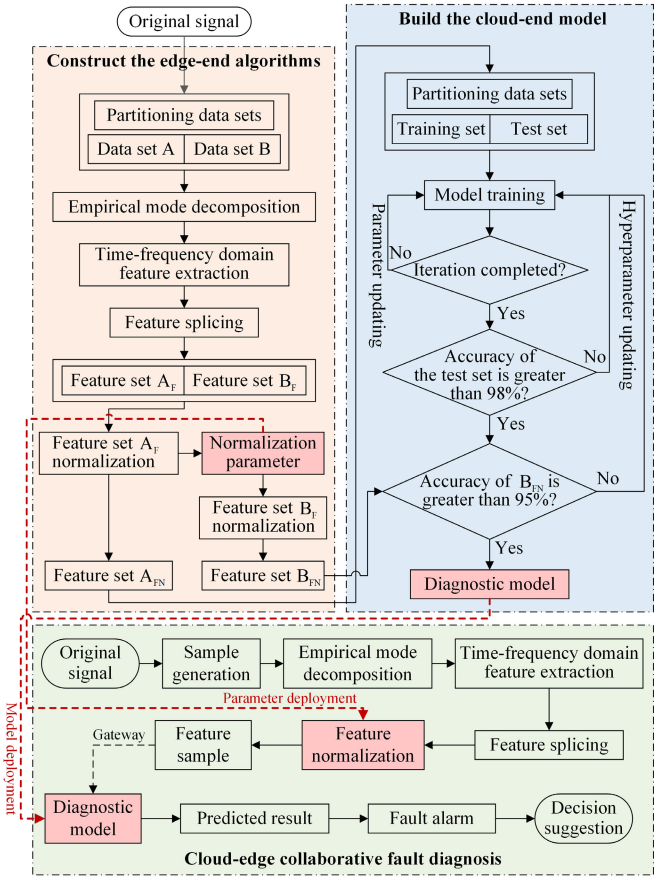


Fig. 2. Flowchart of fault diagnosis algorithm.

conversion units, microprocessors, and wireless transmission modules, in order to realize signal acquisition, edge computing, and data transmission. Signal acquisition is realized by sensors, conditioning circuits, and A/D conversion units. The sensors include piezoelectric sensors (Model 608A11, IMI SENSORS Inc.) and eddy current sensors (OD900503-03-03-50-00, OUDUO Ltd.), which acquire vibration acceleration and displacement signals, respectively. The conditioning circuits mainly consist of coupling capacitors to remove the dc bias from the excitation source required for sensor operation. The A/D conversion unit is mainly composed of an A/D converter chip (AD7606, Analog Devices Inc.) and peripheral circuits to realize the conversion of analog signals to digital signals. The edge computation is realized by a microcontroller unit (MCU; STM32F407, STMicroelectronics Inc.), which mainly completes the execution of signal processing and feature extraction algorithms. The AD7606 and MCU are connected to each other using serial peripheral interface (SPI) communication. The data transmission is implemented using a Wi-Fi module (ESP-01, Ai-Thinker Ltd.), which realizes the bidirectional data sending between the edge end and the cloud end.

B. Cloud-End Scheme

As shown in Fig. 1, the cloud platform (Linkedti Ltd.) consists of data management, a rule engine, and decision suggestions. The data management section stores, recalls, and displays signal waveform results of multiple measuring points from the edge end and can analyze the signal spectrum of the original data and display the data features obtained from the edge end. The rule engine mainly carries out fault diagnosis, fetches feature data and sends them to the classification model for fault inference to get the predicted labels, in which the classification model is a deep learning model trained in advance. The decision suggestion is based on the prediction label obtained by the rule engine to give the fault diagnosis results and then send the disposal decision to the site, thus realizing the cloud–edge collaborative fault diagnosis.

III. Algorithms and Datasets of Cloud–Edge Collaborative Fault Diagnosis

The algorithms in this study are divided into two parts, i.e., signal processing and deep learning. The signal processing includes EMD and time–frequency feature extraction, which is done at the edge end. The deep learning model is 1D-ResNet, which is done at the cloud end. As shown in Fig. 2, it is a flowchart of the cloud–edge collaborative fault diagnosis technology. The original signal is collected at the edge end. The preparation of the feature set is completed by performing EMD and time–frequency domain feature extraction on the original signals, splicing the features, and performing normalization operations. The diagnostic model construction is completed by training the deep learning model using the feature set prepared at the edge end. The edge-end algorithms and the cloud-end model are deployed on the edge device and cloud server, respectively, to realize cloud–edge collaborative fault diagnosis of a rotor-bearing system. It is worth noting that, as indicated by the dotted line in Fig. 2, the normalization parameters used for diagnosis come from the normalization parameters used when the algorithm was built.

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A. Edge-End Signal Processing

1) EMD Algorithm: EMD [27] is an adaptive analysis algorithm for nonlinear, nonstationary time series signals, whose core idea is to decompose the signal into a number of intrinsic mode functions (IMFs) and a residual component. Each IMF represents a different frequency component of the signal, and the residual component represents the trend of the signal.

The EMD of the edge-end signal is as follows:

$$x(t) = \sum_{i=1}^{k} \mathrm{IMF}_i(t) + r_k(t) \tag{1}$$

where $x(t)$ is the original signal; $\mathrm{IMF}_i(t)(i = 1, 2, \ldots, k)$ is the EMD effective component; and $r_k(t)$ is the residual component.

The specific steps of EMD are as follows [27], [28].

Step 1: Find all maximum and minimum values from the original signal.

Step 2: Using the cubic spline interpolation method, the upper envelope $e_{\max}(t)$ is constructed through all local maximum points, and similarly, the lower envelope $e_{\min}(t)$ is constructed through all local minimum points.

Step 3: Calculate the mean values of the upper and lower envelopes to obtain the mean $m_i(t)$

$$m_i(t) = (e_{\max}(t) + e_{\min}(t))/2. \tag{2}$$

Step 4: Extract the IMF and subtract the mean from the original signal to obtain a new signal $h_i(t)$

$$h_i(t) = x(t) - m_i(t). \tag{3}$$

Step 5: Check whether $h_i(t)$ satisfies the IMF termination condition and, if so, $h_i(t)$ is recorded as an IMF component. Otherwise, treat $h_i(t)$ as a new signal and repeat the above steps until the termination condition is satisfied. The modal function $\mathrm{IMF}_i(t)$ is

$$\mathrm{IMF}_i(t) = h_i(t). \tag{4}$$

Step 6: Use the original signal to subtract the modal function to get a new residual component $r_i(t)$

$$r_i(t) = x(t) - \mathrm{IMF}_i(t). \tag{5}$$

Step 7: Replace $x(t)$ in *Step 1* with $r_i(t)$, and repeat the above steps until the second IMF component $\mathrm{IMF}_k(t)$ or the residual component $r_k(t)$ therefore is less than a predetermined value. Alternatively, the EMD process terminates when the residual component $r_k(t)$ is a monotonic function or a constant.

The layer number of EMD and the threshold of the residual component are set in advance at the edge end, and the information of the original signal in different frequency bands is obtained by EMD, which lays a foundation for feature extraction.

2) Feature Extraction Algorithm: The edge-end features include 15 time-domain features and five frequency-domain features [29], where the time-domain metrics are shown in Table I. Frequency-domain feature extraction requires a fast Fourier transform (FFT) of the original signal and then the power spectrum; the power spectrum is used to obtain the frequency-domain features, and the formula is shown in Table II. Time-domain features reflect the changes in the signal over time and can detect the instantaneous changes or shocks in the signal. The frequency-domain features reflect the changes in the signal over the spectrum and can be used to analyze the periodicity of the signal, the harmonic components, and the changes in the frequency components [30]. Considering the influence of working conditions and fault sensitivity, two types of quantitative and dimensionless indicators are selected [31].

TABLE I
TIME-DOMAIN CHARACTERISTIC PARAMETERS AND THEIR FORMULAS

No.	Feature	Formula
1	Maximum	$x_{\max} = \max(x_i)$
2	Maximum absolute	$\lvert x\rvert_{\max} = \max(\lvert x_i\rvert)$
3	Minimum	$x_{\min} = \min(x_i)$
4	Peak-to-Peak	$x_{\mathrm{pp}} = x_{\max} - x_{\min}$
5	Mean	$\overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$
6	Root amplitude	$x_{\mathrm{r}} = \left(\frac{1}{N}\sum_{i=1}^{N}\sqrt{\lvert x_i\rvert}\right)^2$
7	Standard deviation	$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2}$
8	Absolute mean	$\overline{\lvert x\rvert} = \frac{1}{N}\sum_{i=1}^{N}\lvert x_i\rvert$
9	Root mean square	$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}$
10	Kurtosis	$x_{\mathrm{K}} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \overline{x})^4}{\sigma^4}$
11	Skewness	$x_{\mathrm{S}} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \overline{x})^3}{\sigma^3}$
12	Margin factor	$M = \lvert x\rvert_{\max} / x_{\mathrm{r}}$
13	Waveform factor	$W = RMS / \overline{\lvert x\rvert}$
14	Impulse factor	$I = \lvert x\rvert_{\max} / \overline{\lvert x\rvert}$
15	Crest factor	$C = \lvert x\rvert_{\max} / RMS$

The time-domain features in Table I form a vector **TF**

$$\mathbf{TF} = \begin{bmatrix} x_{\max}, \lvert x\rvert_{\max}, x_{\min}, \bar{x}, x_{\mathrm{PP}}, \overline{\lvert x\rvert}, \mathrm{RMS} \\ x_{\mathrm{r}}, \sigma, x_{\mathrm{K}}, x_{\mathrm{S}}, M, W, I, C \end{bmatrix}. \tag{6}$$

The frequency-domain features in Table II form a vector **FF** = [*CF, MF, RMSF, VF, PSE*], which is spliced with **TF** to form the final feature vector **F** = [**TF**, **FF**].

B. Cloud-End CNN Modeling

ResNet as a CNN model is designed to solve the problems of gradient vanishing or exploding and convolutional network degradation due to increasing network depth [32]. In a normal convolutional network, the input passed to the next layer is $H(x) = F(x)$ and is directly fit to $H(x)$. In ResNet, the input passed to the next layer becomes $H(x) = F(x) + x$ and the

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TABLE II
FREQUENCY-DOMAIN CHARACTERISTIC PARAMETERS AND THEIR FORMULAS

No.	Feature	Formula
1	Center frequency	$CF = \frac{\sum_{i=1}^{n} f_i \times ps_i}{\sum_{i=1}^{n} ps_i}$
2	Mean frequency	$MF = \frac{1}{n}\sum_{i=1}^{n} ps_i$
3	RMS frequency	$RMSF = \sqrt{\frac{\sum_{i=1}^{n} f_i^2 \times ps_i}{\sum_{i=1}^{n} ps_i}}$
4	Frequency variance	$VF = \frac{\sum_{i=1}^{n} (f_i - CF)^2 \times ps_i}{\sum_{i=1}^{n} ps_i}$
5	Power spectrum entropy	$PSE = -\sum_{i=1}^{n} \left(\frac{ps_i}{\sum_{i=1}^{n} ps_i} \times \log_2 \frac{ps_i}{\sum_{i=1}^{n} ps_i} \right)$

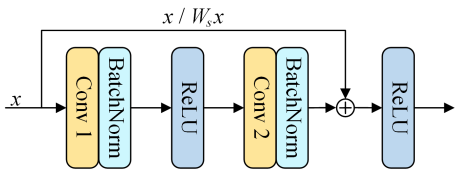


Fig. 3. Structure of residual block.

residuals are fit to $F(x) = H(x) - x$. The residual block does the constant mapping on the one hand, and on the other hand fits the residuals relative to the original network, correcting the bias of the original network without making the network fit the full underlying mapping. The essence of ResNet is that the network learns that the layer parameters are redundant, it can directly choose to skip the redundant layers and the deeper network will not be worse than the shallow network after adding the constant mapping [33]. The structure of the residual block is shown in Fig. 3.

Constant mapping is applied to each stacked layer using residual learning, and the residual block formula is defined as

$$y = F(x, \{W_i\}) + x \tag{7}$$

where x and y are the input and output of the residual block, respectively; and $F(x, \{W_i\})$ is the residual mapping to be learned. If the dimensions of $F(x, \{W_i\})$ and x do not match, they need to be matched by an execution square or a linear projection W_s

$$y = F(x, \{W_i\}) + W_s x. \tag{8}$$

The classification model at the cloud end uses 1D-ResNet. Inputs of 1D-ResNet are 1-D vectors, and the convolution kernel and feature vectors are also in 1-D form [34]. 1D-CNN is more suitable for dealing with 1-D time-series data. 1D-ResNet mainly consists of an input layer, a convolutional layer, a pooling layer, a batch normalization layer, a fully connected layer, and an output layer.

The convolutional layer is the core part of the CNN. It performs a convolution operation on the feature vectors of the previous layer through convolution to obtain corresponding features and uses a nonlinear activation function to construct new output features. The output of each layer is the convolution result of multiple input features, and the formula is

$$y_i^{l+1}(k) = \mathbf{W}_i^l * \mathbf{x}^l(k) + b_i^l \tag{9}$$

where $\mathbf{W}_i^l$ is the weight of the ith convolution kernel in layer l; $\mathbf{x}^l(k)$ is the kth local region feature to be convolved in layer l; b_i^l is the bias of the ith convolution kernel in layer l; $y_i^{l+1}(k)$ is the output of the kth neuron after the ith convolution operation in layer l, i.e., it is used as the input of layer $l+1$; and $*$ denotes the dot-product operation of the convolution kernel and the local region.

After completing the convolution operation, the activation function performs a nonlinear transformation of the convolution output. ReLU function is used as the activation function, denoted as

$$z_i^{l+1}(k) = \max\left\{0, y_i^{l+1}(k)\right\} \tag{10}$$

where $z_i^{l+1}(k)$ is the output of convolution operation and $z_i^{l+1}(k)$ is the activation value of $y_i^{l+1}(k)$.

The pooling layer is used to reduce the size of the feature vectors, and computational complexity and prevent overfitting. The input feature vectors are downsampled to extract the main features. Maximum pooling is used, denoted as

$$p_i^k = \max x_{ab}^k (a, b \in \mathbf{R}_i) \tag{11}$$

where $\mathbf{R}_i$ is the pooling region, x_{ab}^k is the neurons in the pooling region, and p_i^k is the kth feature mapping output of the pooling operation.

The batch normalization layer is used to speed up the training process of the network and improve its stability by normalizing the inputs of each layer to suppress the problem of vanishing or exploding gradients. The batch normalization is denoted as

$$q_i^l = \gamma^l \left(\frac{x_i^l - \mu^l}{\sqrt{(\sigma^l)^2 + \varepsilon}} \right) + \beta^l \tag{12}$$

where x_i^l is the input; μ^l and $(\sigma^l)^2$ are the mean and variance of the input, respectively; γ^l and β^l are the scaling and translation parameters, respectively; q_i^l is the batch normalized output; and ε is a very small constant.

The fully connected layer transforms the output of the last pooling layer and maps the input to get a vector. The fully connected layer is

$$u^{l+1}(j) = f\left(W_{ij}^l(m) z_i^l(m) + b_i^l\right) \tag{13}$$

where $W_{ij}^l(m)$ is the weight value between the mth neuron of the ith channel of the lth layer and the output neuron of the jth channel of layer $l+1$ is the output value of the mth neuron of the ith channel of the previous layer, $z_i^l(m)$ is the bias of the network, $u^{l+1}(j)$ is the output value of the mth neuron of the jth channel, and $f(\cdot)$ is the activation function.

The output layer uses softmax as the classifier. The structure of 1D-ResNet is shown in Fig. 4. Experimental validation of fault diagnosis algorithms.

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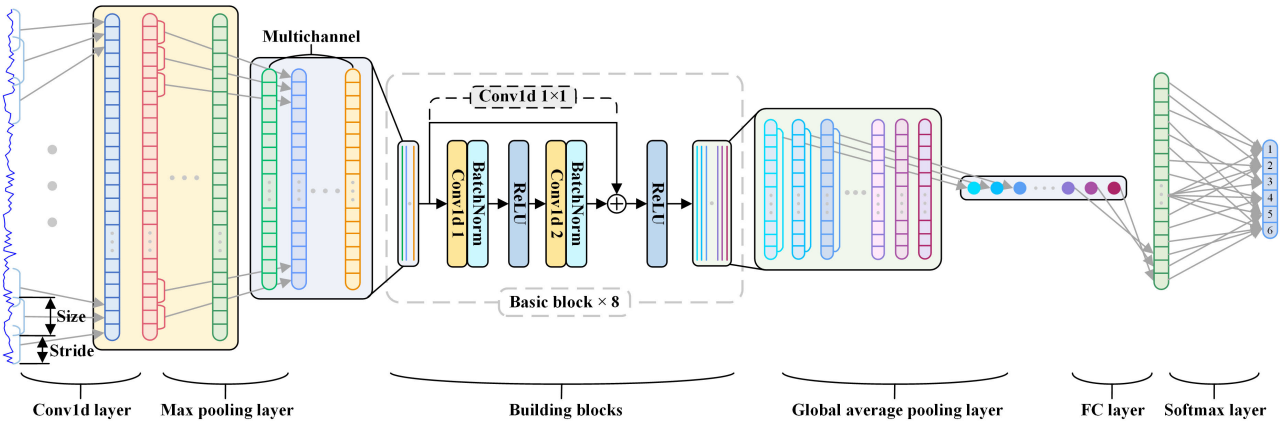


Fig. 4. Structure of 1D-ResNet.

TABLE III
MAIN PARAMETERS OF THE EXPERIMENTAL PLATFORM

Element	Main parameters
Drive system	Speed 0–3000 r/min, 3-phase motor
Shaft system	NSK-6205 bearing, shaft with disc
Gearbox	Speed ratio 4:1
Loading device	Torque range 0–50 N·m

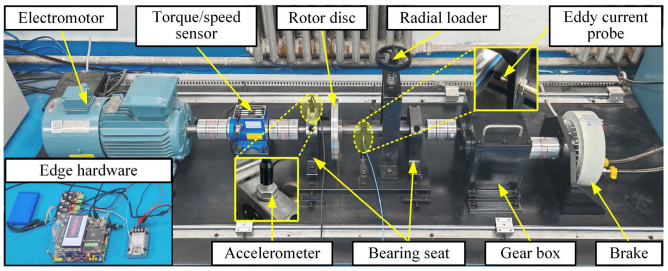


Fig. 5. Fault simulation platform of rotor-bearing system.

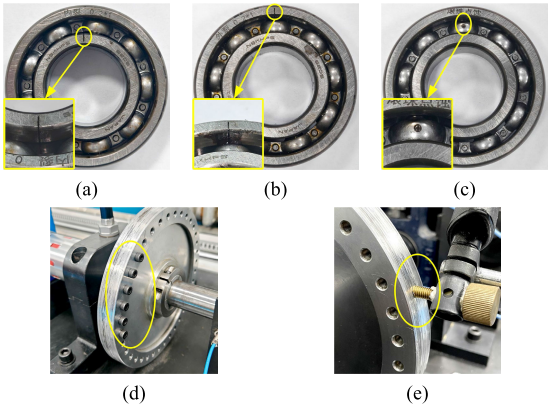


Fig. 6. Fault scheme of rotor-bearing system. (a) Inner ring fault (BI). (b) Outer ring fault (BO). (c) Ball fault (BB). (d) RI. (e) Rotor rub-impact (RR).

TABLE IV
MAIN PARAMETERS OF DATA ACQUISITION SYSTEM

Device	Main parameters	Values
Accelerometer	Measurement range	±50 g
	Sensitivity	0.089 V/g
	Non-linearity	±1%
Eddy current probe	Measurement range	0.5–1.5 mm
	Sensitivity	4 V/mm
	Non-linearity	±0.5%
A/D converter	Conversion accuracy	16 bit
	Maximum sampling rate	200 kHz
	Dynamic range	90 dB

C. Experimental Platform Development

The experimental platform used is shown in Fig. 5, and it consists of a motor, a shaft system, a gearbox, a brake, and a control cabinet, and the main parameters are shown in Table III. As shown in Fig. 5, the faults simulated are located in the shaft system, in which the rolling bearing fault is set at the bearing near the drive end and the rotor fault is set on the disk. As shown in Fig. 6(a)–(c), the fault types of the rotor-bearing system include inner ring fault, outer ring fault, and rolling element fault of rolling bearing, as well as the normal state. Inner and outer ring faults are fabricated as rectangular grooves across the raceways, and rolling element fault is produced by laser ablation. Rotor faults include rotor imbalance (RI) and rub-impact, where RI is generated by screwing six screws of the same material and size in the disk, and rub-impact is realized by touching between the copper screws on the stationary holder and the rotating disk, as shown in Fig. 6(d) and (e).

The parameters of the data acquisition system are shown in Table IV. The sampling rate is set to 6000 Hz, the number of sampling points is set to 102 400, and the sampling duration is 17.07 s.

D. Dataset Construction

In the experiment, the motor speeds were 700 and 1400 r/min, and no load and 25 N·m load were simulated by setting 0 and 0.5 A currents to the magnetic brake. The typical data collected are shown in Fig. 7. Five groups of data are collected for each fault type and each working condition, and the length of a single group of data is 2048, so the number of samples for each fault type and each working condition is 250, and the total number of samples is $2 \times 6 \times 250 = 3000$.

This experiment uses the load-related input strategy and unlabeled load output strategy. The load types include no load and a load of 25 N·m. The 3000 samples were divided

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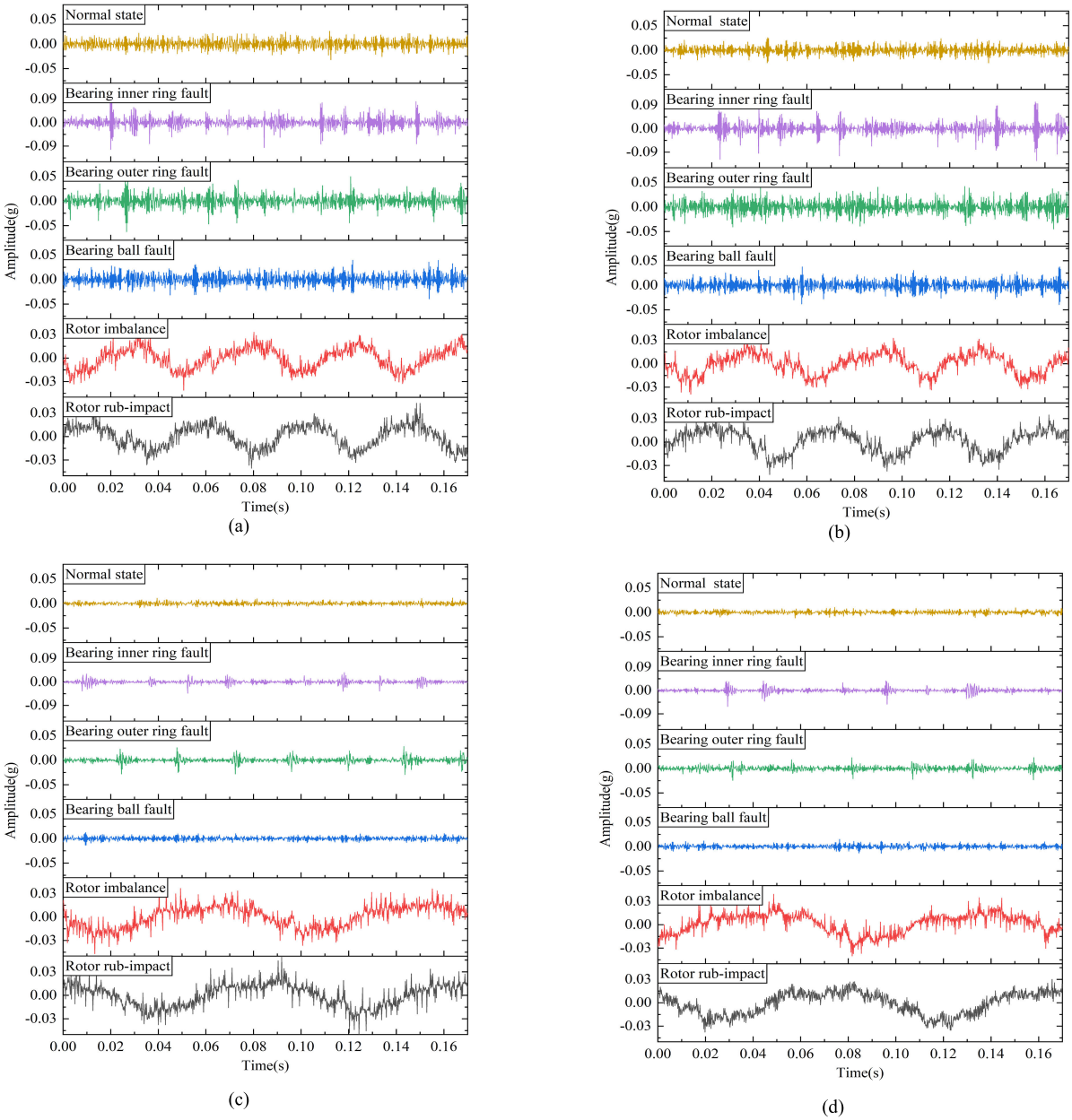


Fig. 7. Time-domain waveform of original signal. (a) No load and 1400 r/min. (b) 25 N·m and 1400 r/min. (c) No load and 700 r/min. (d) 25 N·m and 700 r/min.

into datasets A and B. Dataset A contains 400 samples for each fault type, totaling 2400 samples. Dataset B contains 100 samples of each fault type, totaling 600 samples. Divide the samples in dataset A into a training set and a test set for model training and test. Use dataset B for the test only to simulate the data collected at the edge end.

IV. Validation of Cloud–Edge Collaborative Fault Diagnosis Method

A. Edge-End Signal Processing

When making feature samples, the original data are first decomposed by EMD to obtain multiple IMF components; then, the time–frequency features are extracted sequentially from each component, and finally, the features are spliced after being spread out in the order of components to obtain the feature vector. If the number of decomposition layers is n, the length of the final feature vector is n times the length of the vector obtained by direct feature extraction of original data.

1) EMD of Tested Data: Taking the bearing inner ring fault at 1400 r/min as an example, the number of decomposition layers is set to 5, the IMF components obtained are shown in Fig. 8(a), and Fig. 8(b) shows the FFT results of the original signal and its components. From Fig. 8(a), it can be seen that the time-domain waveform of IMF1 is close to that of the original signal, and with the increase of the number of decomposition layers, the time-domain waveform of the component gradually tends to flatten. From Fig. 8(b), it can be seen that with the increase of the number of decomposition layers, the frequency bands in the spectrum of the component change from high frequency to low frequency, and the bandwidths

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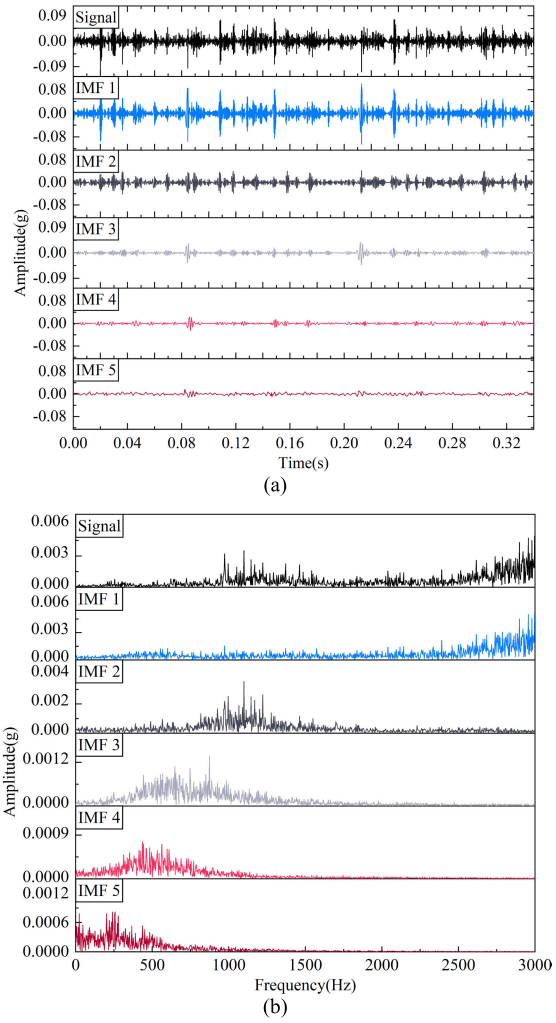


Fig. 8. EMD results for bearing inner ring fault. (a) Time-domain waveform of IMF components. (b) FFT spectrum for IMF components.

add up to be approximately equal to the total width of the frequency bands of the original signal.

2) Feature Extraction and Sample Construction: After the above EMD, the IMF components are obtained, and 20 time–frequency feature values are extracted for these components in turn. The feature vector extracted for each component can be denoted as $\mathbf{F}_{\mathrm{IMF}i}$, i.e.,

$$\mathbf{F}_{\mathrm{IMF}i} = \begin{bmatrix} x^i_{\max}, |x|^i_{\max}, x^i_{\min}, \bar{x}^i, x^i_{\mathrm{PP}}, \overline{|x|}^i, \mathrm{RMS}^i \\ x^i_{\mathrm{r}}, \sigma^i, x^i_{\mathrm{K}}, x^i_{\mathrm{S}}, M^i, W^i, I^i, C^i, CF^i \\ MF^i, RMSF^i, VF^i, PSE^i \end{bmatrix}. \quad (14)$$

The feature vectors extracted from all IMF components are spliced sequentially, and then, the feature vector $\mathbf{F}_{\mathrm{Sample}}$ for each sample can be obtained, i.e.,

$$\mathbf{F}_{\mathrm{Sample}} = [\mathbf{F}_{\mathrm{IMF1}}, \mathbf{F}_{\mathrm{IMF2}}, \ldots, \mathbf{F}_{\mathrm{IMF}n}] \quad (15)$$

where n is the number of EMD layers.

According to the above, five-layer EMD, time–frequency feature extraction, and feature vector splicing are performed sequentially on the samples of six states in datasets A and B; finally, feature sample sets $\mathrm{A_F}$ and $\mathrm{B_F}$ are obtained.

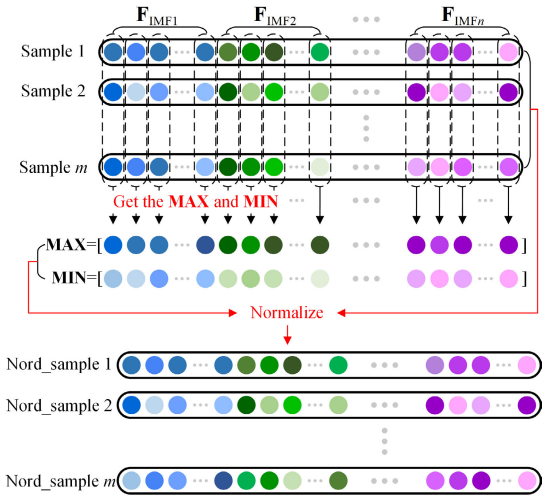


Fig. 9. Schematic of normalization operation.

In the feature datasets $\mathrm{A_F}$ and $\mathrm{B_F}$, each feature sample vector contains multiple features, and the different features differ in dimension and order of magnitude. If the feature data are directly fed into the model for training, it will result in certain features having too much or too little influence on the training process, thus affecting the convergence speed and stability of the model. If each feature sample vector is normalized according to its own, the differences between different features cannot be eliminated. The normalization according to feature dimensions can scale all features to the same magnitude so that the model can better learn the relative importance of each feature, and at the same time, it can inhibit the overfitting of the model to certain features to a certain extent, and improve the generalization of the model.

In this study, the feature samples are maximum–minimum normalized according to the feature dimensions, such as

$$F_{\mathrm{Nord}}(j) = \frac{F_{\mathrm{Sample}}(j) - \max(F_{\mathrm{Sample}})}{\max(F_{\mathrm{Sample}}) - \min(F_{\mathrm{Sample}})} \quad (16)$$

where $j = 1, 2, \ldots, 20 \times n$, in which n is the layer number of EMD, and $F_{\mathrm{Nord}}(j)$ is the normalized feature sample.

As shown in Fig. 9, taking the dataset $\mathrm{A_F}$ as an example, the feature samples (Sample $1, 2, \ldots, m$, where $m = 2400$) are arranged in the way shown in the figure, and the maximum and minimum values of features in each column are taken to form the maximum value vector **MAX** and the minimum value vector **MIN**. Then, the feature samples of the dataset $\mathrm{A_F}$ are normalized using the maximum and minimum value vectors to obtain new feature samples (Nord_sample $1, 2, \ldots, m$, where $m = 2400$). For the dataset $\mathrm{B_F}$, because of the small number of samples, $\mathrm{B_F}$ is normalized with the normalization parameters **MAX** and **MIN** of the dataset $\mathrm{A_F}$ in order to ensure consistency with the feature distribution of the dataset $\mathrm{A_F}$ and that the feature space of the subsequent test model is uniform. After datasets $\mathrm{A_F}$ and $\mathrm{B_F}$ are normalized as described above, datasets $\mathrm{A_{FN}}$ and $\mathrm{B_{FN}}$ are finally obtained for subsequent model training and test.

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TABLE V
PARAMETERS OF IMPROVED 1D-RESNET MODEL

No.	Layer	Output (C, H)	Parameters
-	Input	(1,120)	-
1	Conv	(64,60)	512
2	MaxPool	(64,30)	0
3	Basic block1	(64,30)	24960
4	Basic block2	(128,15)	82816
5	Basic block3	(256,8)	329472
6	AdaptiveAvgPool	(256,1)	0
7	Linear	6	1542

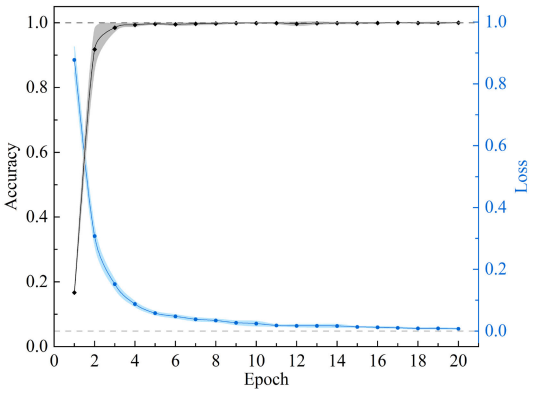


Fig. 10. Accuracy and loss error of model training.

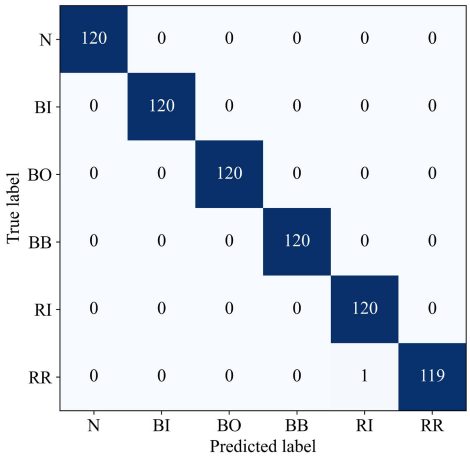


Fig. 11. Confusion matrix of test set.

TABLE VI
MODEL EVALUATION RESULTS

State	Accuracy	Precision		Recall		F1-score	
		Value	Avg	Value	Avg	Value	Avg
N	**0.999**	1.000	**0.999**	1.000	**0.999**	1.000	**0.999**
BI		1.000		1.000		1.000	
BO		1.000		1.000		1.000	
BB		1.000		1.000		1.000	
RI		**0.992**		1.000		**0.996**	
RR		1.000		**0.992**		**0.996**	

B. Cloud-End CNN Modeling

In order to ensure the accuracy and efficiency of fault diagnosis, 1D-ResNet18 is used as the base classification model. Since the time–frequency feature extraction has been performed on the original signal, the number of convolutional layers is reduced for the original 1D-ResNet18. It is found that the model size is reduced from 14.80 to 1.79 MB while reducing the convolutional layers, and the parameters of the improved model are shown in Table V.

The dataset A_{FN} is divided into a training set and test set according to 7:3, and the number of samples is 1680 and 720, respectively, to train the model. First, different hyperparameter values are taken to train the model, and a set of more appropriate hyperparameters is determined by combining the stability and convergence speed of the training process and the accuracy of the test set. After many times of parameter tuning, the cross entropy is finally determined as the loss function, the optimizer is Adam, the learning rate is 0.0001, the batch size is 64, and the number of training epochs is 20. Then, the model is trained for five times using the determined hyperparameters, and the error bands of the accuracy and loss are shown in Fig. 10. It can be seen that the accuracy values of five training epochs have a small fluctuation at the beginning and stabilizes to 100% at the later stage, and the loss values have a small fluctuation in general, and finally stabilizes to 0. The average standard deviation of the accuracy and loss is calculated to be 0.00522 and 0.00752, which makes the model training more stable.

The final epoch of the model for each training is tested using the test set with an average accuracy of 99.9%. Meanwhile, the model is comprehensively evaluated using the confusion matrix, precision, recall, and F1-score, as shown in Fig. 11 and Table VI. From the confusion matrix, it can be seen that only one rotor rub-impact sample is misclassified as RI, and all other category samples are classified correctly. From the evaluation metrics in Table VI, the precision, recall, and F1-score for each category are very close to 100%. Looking at the metrics of each category, for N, BI, BO, and BB, the model reaches 100% in recall, precision, and F1-score. For RI, the recall is 100% but the precision is 99.2%, indicating that a small number of samples are misclassified as other categories. For RR, the precision is 100% but the recall is 99.2%, indicating that a small number of samples in this category are misclassified as other categories. It can be seen that the overall performance of the proposed model in this work is excellent on the task of classifying the state data of the rotor-bearing system.

C. Comparative Analysis of Model Settings at 1400 r/min

Using dataset B_{FN} to simulate actually collected data, the sample length, layer number of EMD, and deep learning model are varied to set up different schemes for comparative experiments. The different schemes are evaluated by training curves, confusion matrix, and t-SNE results. Here, the hyperparameter values used for the model training are kept the same as in Section IV-B.

In the comparative experiments, the sample length is set to 1024 and 2048, the layer number of EMD is set to 4, 5, and 6, and the deep learning model is the original 1D-ResNet18 and its improved model. The initial settings are 1024, 4, and

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TABLE VII
DESIGN OF COMPARATIVE EXPERIMENTS

No.	Sample length		EMD layers			Model(1D-CNN)	
	1024	2048	4	5	6	ResNet18	Proposed
1	√		√			√	
2		√	√			√	
3		√		√		√	
4		√			√	√	
5		√		√			√

TABLE VIII
ACCURACY VALUES OF COMPARATIVE EXPERIMENTS

No.	Accuracy						
	1	2	3	4	5	Avg	Std (10^{-2})
1	0.980	0.957	0.968	0.980	0.983	0.974	1.089
2	0.993	0.995	0.983	0.990	0.980	0.988	0.645
3	0.998	0.997	0.998	0.993	0.998	0.997	0.217
4	0.995	0.990	0.997	0.997	0.993	0.994	0.254
5	**0.998**	**1.000**	**1.000**	**0.998**	**1.000**	**0.999**	**0.093**

1D-ResNet18. A layer-by-layer approach is taken, i.e., the better result of the previous step is retained for use in the next step, and a total of five different scenarios are set up, as shown in Table VII. The diagnostic results of the five scenarios for the sample B_{FN} are shown in Table VIII. In each case, five trainings are performed and the mean (Avg) and standard deviation (Std) are calculated.

In Table VIII, the average accuracy of No.2 is 1.4% higher than that of No.1, and the standard deviation of the accuracy is lower, i.e., the model stability is improved, which proves that when the sample length is 2048; the data contains more information, and the feature differences between different categories are amplified after feature extraction, while the feature similarity among samples within each category is increased. Comparing No.2–No.4, the average accuracy of five-layer EMD is higher compared to those of 4 and 6, reaching 99.7%, with the lower standard deviation and better stability, proving that the feature dissimilarity between different categories is amplified after five-layer EMD. If the decomposition is insufficient, the differences between different categories will not be sufficiently amplified, and the similarity within categories will not be adequately reflected. If the decomposition is excessive, the components at the back will basically tend to be linear, amplifying the similarity between different categories and resulting in a decrease in the accuracy of classification. Comparing No.3 and No.5, the final average accuracy and stability are further improved.

To further explain five comparative experiments, the training error bands of each experiment are given, as shown in Fig. 12.

From Fig. 12(a)–(d), it can be seen that No.1–No.4 have large fluctuations in accuracy and loss during model training, and the model stability is poor, which can easily lead to a decrease in the model generalization performance, which is not conducive to site applications. From Fig. 12(e), it can be seen that the training accuracy and loss of No.5 are stable, indicating that No.5 has the most reasonable matching of each

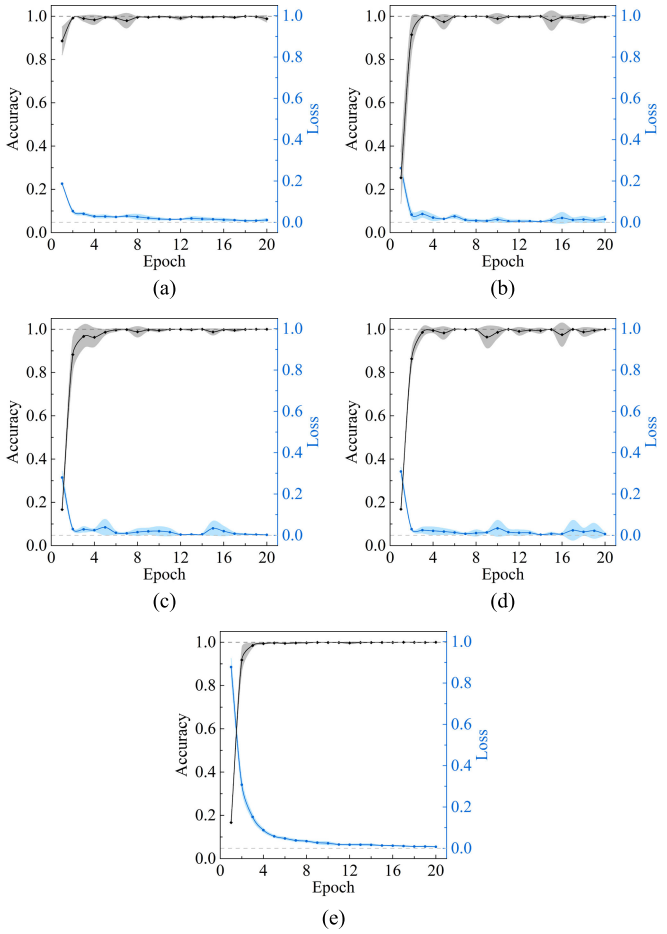


Fig. 12. Error bands for the model training process for the five sets of experiments. (a) No.1. (b) No.2. (c) No.3. (d) No.4. (e) No.5.

parameter and high model stability, which is suitable for use in practice.

The classification results for the dataset B_{FN} are illustrated by selecting the confusion matrix with the worst results in each experiment, as shown in Fig. 13. Misclassification occurs for the normal state, ball fault, and RI in Fig. 13(a), with the worst misclassification occurring for RI. A small amount of misclassification occurs for RI in Fig. 13(b). There are individual samples misclassified in Fig. 13(c)–(e), which are generally better. From Fig. 13(a), (b), and (d), it can be seen that there are some samples of RI that are misclassified as rotor rub-impact. From the data, the original data of the two categories are more similar. From the model, rotor rub-impact is given more weight. From Fig. 13(a) and (c) to (e), it can be seen that there is a misclassification between the normal state and ball fault, which indicates that there is a higher similarity between data of the two categories, and the model is prone to errors when recognizing the two categories of fault. From the confusion matrix, Fig. 13(e) has the best performance, which again proves that the parameter setting of No.5 is the most reasonable.

In order to reveal the feature distributions of different models, the feature distributions at three key positions are downscaled to give t-SNE visualization results, as shown in Fig. 14. No.X-A shows the feature distribution of the original

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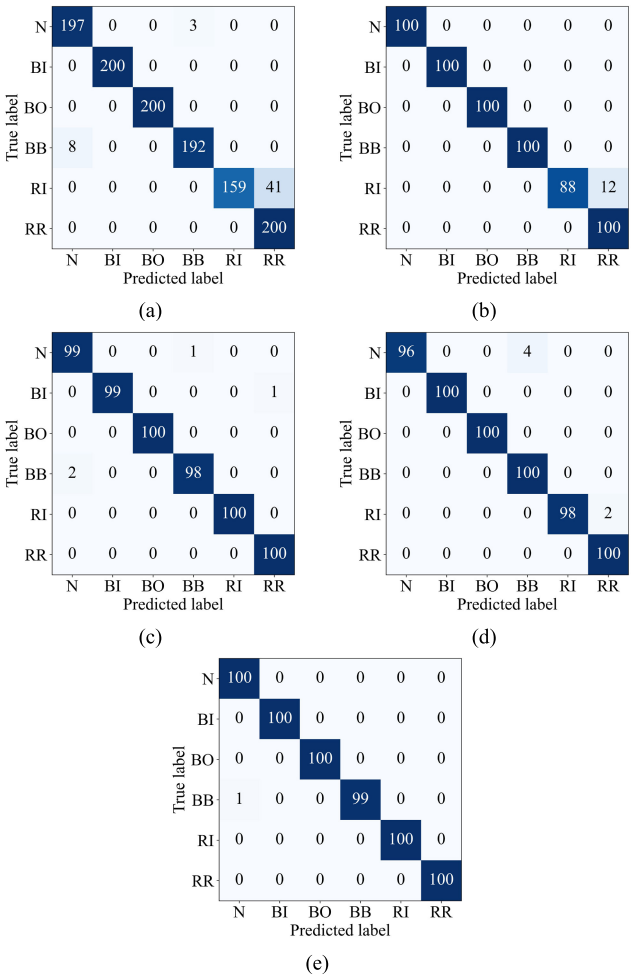


Fig. 13. Confusion matrix of different models for the five sets of experiments. (a) No.1. (b) No.2. (c) No.3. (d) No.4. (e) No.5.

data input layer, and the feature distribution of No.1 is different from the other models. No.X-B shows the feature distribution of the model input layer, i.e., the sample feature distribution after signal processing and feature extraction. No.X-C shows the model linear input layer feature distribution, i.e., the feature distribution after convolutional feature extraction. Compared to No.X-A with No.X-B, the initial separation of features can be completed, which verifies the effectiveness of signal processing and feature extraction. Compared to No.X-B with No.X-C, the convolutional layer carries out the secondary feature extraction, which finally separates each category. The results of No.2–No.5 are better than No.1, and there is no obvious overlapping of their features, while the feature distributions of the normal state, ball fault, RI, and rub-impact in No.1 overlap. Compared to No.2-C with No.4-C, it can be seen that there are individual samples with misclassification and intersecting feature distributions, while from No.5-C, it can be seen that the features are correctly separated, and the classification effect is the best.

D. Comparative Analysis of Diagnostic Models at 1400 r/min

The proposed method is compared with other deep neural networks widely used in fault diagnosis. The input

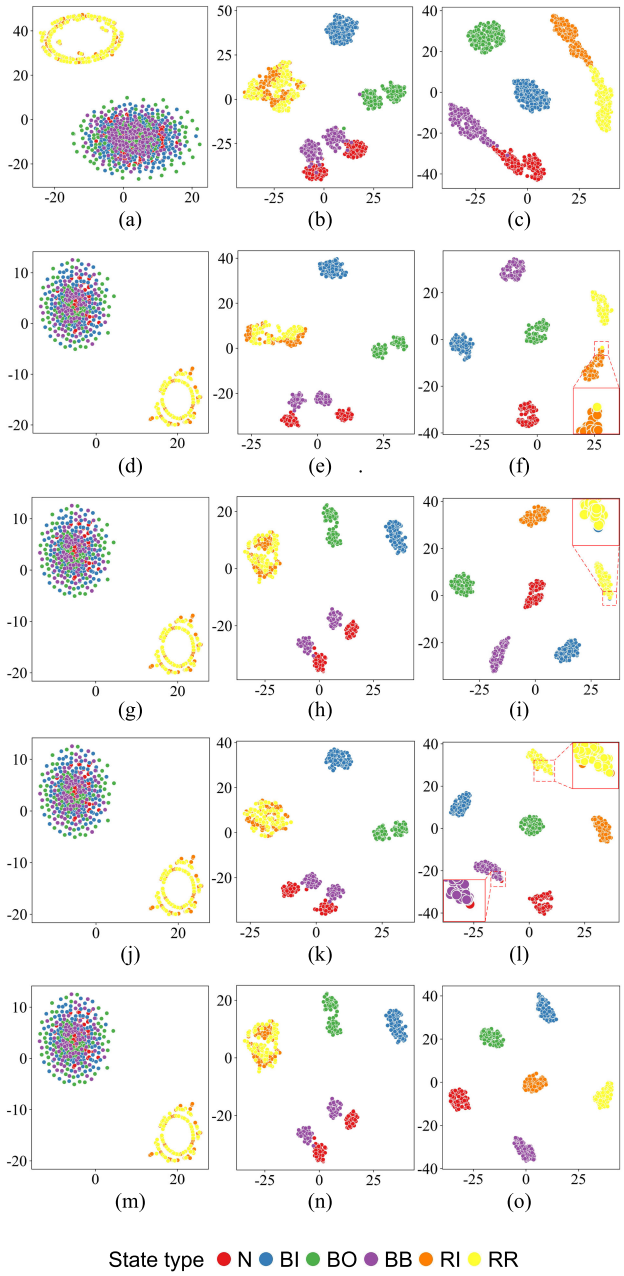


Fig. 14. Feature distributions using t-SNE: “A” represents the original data input layer, “B” represents the model input layer, and “C” represents the model linear input layer. (a) No.1-A. (b) No.1-B. (c) No.1-C. (d) No.2-A. (e) No.2-B. (f) No.2-C. (g) No.3-A. (h) No.3-B. (i) No.3-C. (j) No.4-A. (k) No.4-B. (l) No.4-C. (m) No.5-A. (n) No.5-B. (o) No.5-C.

samples of different methods are all feature samples obtained by signal processing and feature extraction of the original data collected by the edge end. Seven different CNNs, i.e., LeNet-5 [35], AlexNet [36], VGG-19 [36], ResNet-18 [32], DenseNet-121 [37], ShuffleNet-V2 [20], and MobileNet-V3 [21], are selected to compare their classification capabilities on the same dataset. In the training, the learning rate is set to 0.0001, the optimizer is Adam, the batch size is 64, and the number of training epochs is 20. Each model is trained five times, and the model obtained after the last epoch is selected for each training, and its performance is verified on

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TABLE IX
PERFORMANCE OF DIFFERENT MODELS AT 1400 r/min

Model	Accuracy		Parameters	FLOPs	Model
	Avg	Std(10^{-2})	(M)	(M)	size(MB)
LeNet	0.760	9.164	0.572	0.812	2.180
AlexNet	0.978	1.496	3.408	6.918	13.000
VGG	0.975	0.952	11.938	79.645	45.540
ResNet	0.997	0.217	3.850	20.294	14.690
DenseNet	0.987	1.171	7.432	48.735	28.350
ShuffleNet	0.960	1.577	0.974	4.739	3.720
MoblileNet	0.978	1.638	2.820	4.561	10.760
Proposed	**0.999**	**0.093**	**0.439**	**4.040**	**1.680**

the dataset B_{FN}. Meantime, the parameters, FLOPs, model size, and average test accuracy are analyzed.

The results are shown in Table IX. Except for LeNet-5, the average accuracies of the other models are all above 95%. On the one hand, it is verified that the LeNet-5 model with a relatively simple structure cannot perform sufficient feature extraction, resulting in a lower accuracy. On the other hand, it proves that the original data have obtained better features after signal processing and feature extraction at the edge end. The proposed method has the best average accuracy, as well as the lowest parameter and model size. Its performance is also the most stable, with a standard deviation of 0.00093. Although the ResNet-18 model has comparable accuracy to the proposed method, its parameters, FLOPs, and model size are higher than those of the proposed method.

E. Model Generalization Verification at a Lower Speed

The proposed method is further verified under a low speed of 700 r/min and two loads of 0 and 25 N·m. The dataset partitions, sample lengths, and EMD decomposition layers are the same as those at 1400 r/min. In training, the learning rate is set to 0.001, the optimizer is Adam, the batch size is 256, the number of training epochs is 100, and the model is trained five times. The average test accuracy reaches 94.8%, which is a decrease of 5.1% compared to that at 1400 r/min, and the standard deviation is 0.00380.

The training error band and confusion matrix are shown in Fig. 15(a) and (b), respectively. When the speed drops to 700 r/min, it can be seen from Fig. 15(a) that the convergence speed of the model slows down, and more training epochs are required to make the accuracy and loss converge. The early training curve appears to fluctuate, but the model eventually converges stably, with the accuracy approaching 100% and the loss approaching 0. As can be seen from Fig. 15(b), the main categories that cause the model accuracy to decrease are RI and RR, with most misclassified cases occurring between these two, and a few misclassified cases occurring for N. The t-SNE-based visualization results of the feature distributions at the three key points are shown in Fig. 16(a)–(c). It can be seen from Fig. 16(a) and (b) that the edge end has a weaker feature extraction ability in distinguishing the features of categories N and BB than that at 1400 r/min. It can be seen from Fig. 16(b) and (c) that the feature distributions of

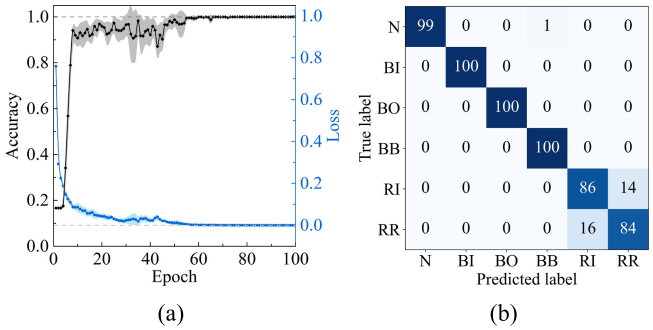


Fig. 15. Model performance at 700 r/min. (a) Training error band. (b) Confusion matrix.

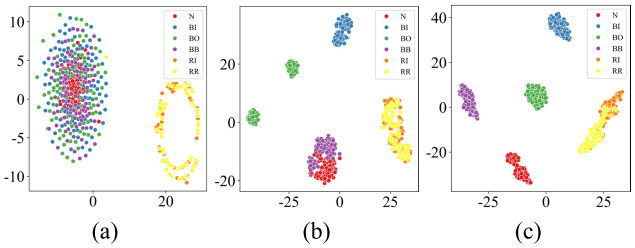


Fig. 16. Feature distributions at 700 r/min using t-SNE. (a) Original data input layer. (b) Model input layer. (c) Model linear input layer.

categories RI and RR overlap, while the other categories are separated.

The results above at a lower speed show that the accuracy of the proposed method varies. At 700 r/min, the mechanical fault-induced vibration is weakened, and the feature extraction effect is not as obvious as that at 1400 r/min, resulting in a decrease in the model accuracy. Meanwhile, the model can still converge to a high accuracy at 700 r/min, which proves that the proposed method has a certain generalization capability at different speeds.

V. VALIDATION OF CLOUD–EDGE COLLABORATIVE SYSTEM

A. Edge-End Unit Verification

First, the edge-end algorithms are compiled with C code on STM32Cube. Second, the edge-end hardware is developed, and the code is burned into the hardware and debugged to the ideal effect. In order to view the progress of edge computation, a liquid crystal display (LCD) is used. As shown in Fig. 17, LCD displays the numerical results, original signal waveform, and feature sample.

B. Cloud-End Platform Verification

For the cloud-end test, first, the fault diagnosis rule shown in Fig. 18(a) is established, where the labeled data module is the feature sample uploaded from the edge end, the classification model module is the source file mapping of the proposed model, and the alarm module is the display of the fault diagnosis results. The rule for cloud-end fault diagnosis is as follows. The labeled data uploaded from the edge end are sent to the classification model trained offline, and the fault diagnosis alarm results are given by running the classification

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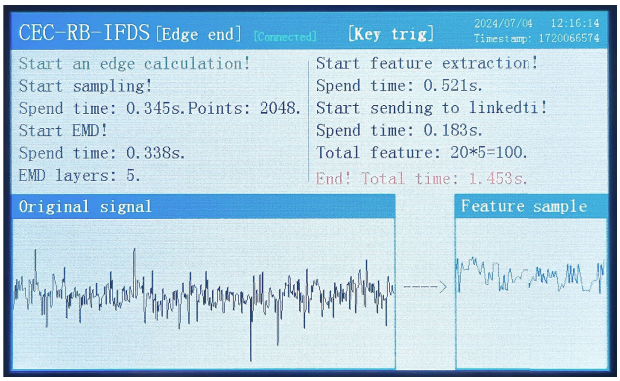


Fig. 17. Test results from LCD at edge end.

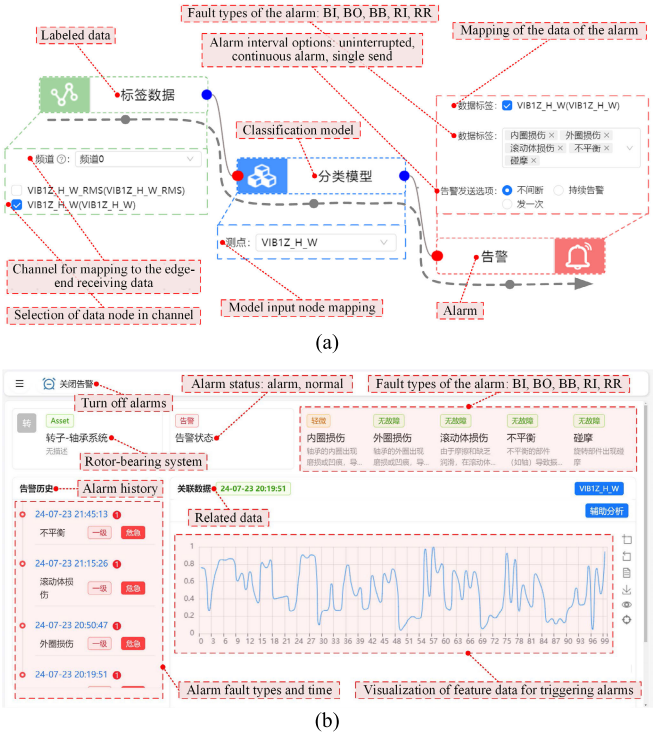


Fig. 18. Fault diagnosis rule and results at cloud platform. (a) Fault diagnosis rule at cloud platform. (b) Fault diagnosis results at cloud platform.

model on the test data. The alarm page is shown in Fig. 18(b), which can display feature data and fault diagnosis results, and give corresponding alarms based on the fault diagnosis results.

C. Time-Consuming Test

Since the time-consuming process of cloud–edge collaborative computing is mainly at the edge end, the time-consuming process of the No.1–No.5 tests is tested. The time-consuming procedures of the five models at different stages and the total time-consuming procedures are shown in Fig. 19. The edge computing process of No.3 and No.5 is the same, so only one test is performed. T1 is the time taken by the MCU to control the AD7606 to complete data acquisition, T2 is the time taken to perform EMD processing, T3 is the time taken for time–frequency feature extraction and normalization, and T4 is the time taken to package the feature data into MQTT

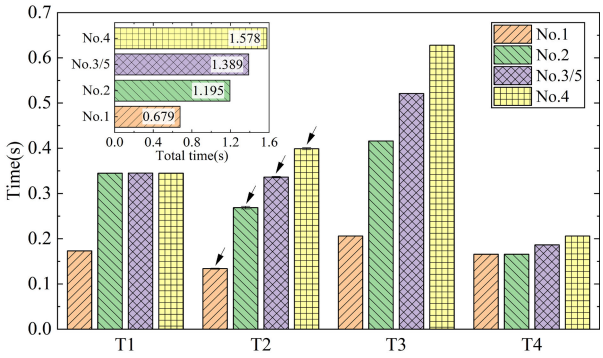


Fig. 19. Time-consuming results of different models.

packets and perform the sending operation. The total time taken for each model is the sum of T1–T4. Overall, the time taken for the five models at different stages is stable. T1 shows that the time taken for 2048 sampling points is twice that taken for 1024 sampling points. T2 shows that the increase in the number of EMD decomposition layers increases the time taken by the program. T3 shows that the increase in IMF components increases the time taken for feature extraction. T4 shows that the increase in the amount of packaged data increases the time taken by the program. It can be seen that No.1 has the least total time consumption, and No.4 has the most. In general, the total time consumption of the five models is not much different, and the fault diagnosis results can be obtained in about 1 s. Therefore, combined with the fault diagnosis accuracy, No.5 is selected as the final model.

D. Anti-Noise Test

In practice, the engineering signal is affected by background noise, so the anti-noise performance of the fault diagnosis method needs to be tested. For the original signal, Gaussian white noise with an SNR of 5 to −30 dB, a mean of 0, and a variance of 1 is added sequentially. Five trainings are performed using dataset A, and the average accuracy of the test using dataset B is used as the evaluation metric. The results are shown in Fig. 20. Overall, the average accuracy gradually decreases as the noise intensity increases, and there is a small fluctuation after adding noise. The average accuracy at an SNR of 5 dB is 1.0% lower than that of the noise not added (NA). When the noise intensity increases from 0 to −10 dB, the accuracy rate decreases slightly. When the noise intensity increases from −15 to −30 dB, the accuracy rate decreases significantly. The test accuracy at −30 dB is 90.3%, meaning that 58 of the 600 samples in dataset B are misclassified. At −10-dB noise intensity, the average accuracy remained above 95.0%. If the actual noise is greater, the signal should be filtered before using the algorithms proposed.

E. Edge-End Energy Consumption Test

The energy consumption of the edge-end device is tested. The power supply uses 24 V DC, and the measured result is 23.97 V. By measuring the input current with an ammeter in series at the power supply input, the input power of the edge-end device during operation is calculated to be 2.924 W.

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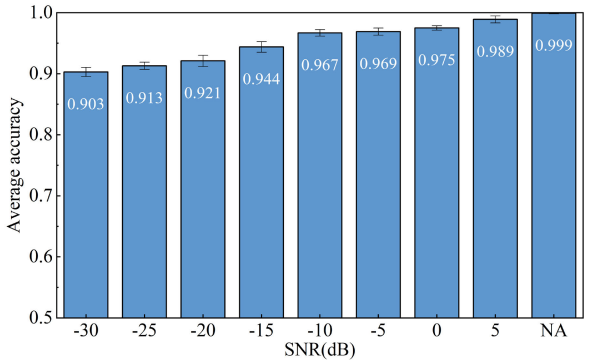


Fig. 20. Test average accuracy of samples with noise.

TABLE X
POWER CONSUMPTION OF EDGE-END DEVICE (UNIT: W)

MCU and ADC	DC-DC	WIFI	Sensor	LCD	Edge computing	Total
1.079	0.671	0.120	0.120	0.838	0.096	2.924

Second, by controlling whether each module is working, the power consumption of each module and the power consumption of the edge-end device when performing edge computing once are measured, as shown in Table X. The power consumption of the edge-end device is mainly concentrated on the MCU, ADC, DC–DC, and LCD. If the power consumption of the edge-end device is controlled, the power consumption of these modules can be controlled. For example, a low-power mode can be used for the MCU, a step-down module with a higher power conversion efficiency can be used, and the LCD can be manually woken up when needed.

VI. CONCLUSION

1) A cloud–edge cooperative system and a series of data processing algorithms are developed, and the combination of the two realizes the efficient and accurate fault diagnosis of the rotor-bearing system.
2) A set of edge-end devices includes embedded hardware and an onboard signal processing program, and a cloud-end platform can carry out deep learning-based fault diagnosis and alarm.
3) EMD, feature extraction, and normalization algorithms are used in the embedded device, and an improved 1D-ResNet algorithm is used in the cloud platforms.
4) The experiment realizes six-state category recognition at different conditions, with an accuracy of 99.9% at 1400 r/min and 94.8% at 700 r/min. The time required for fault diagnosis is about 1.4 s.

Although the proposed cloud–edge collaborative fault diagnosis method and system have achieved excellent effects, there is still a need to strengthen research on multisource heterogeneous data fusion, fine diagnosis under complex working conditions, data transmission security, and system popularization and optimization.

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